

Weekly Report

What I worked on this week

Built all programs in the project dir

Created a script to run and profile ABINIT

I created a to run and profile ABINIT, meant to be the blueprint for the other programs

ABINIT profile I ran ABINIT on its 0th test case on 2 nodes with 16 MPI tasks. I set PEAK to profile memory, BLAS, LAPACK, and FFTW.

Library usage report

```
python ../../peak_analysis_v2.py peak_stats-p496734.csv -mem peak_mem-
p496734.csv
```

Loading memory log from: peak_mem-p496734.csv

Memory graphs saved to: peak_mem-p496734_graphs.png

=== Memory Log Summary ===

Total events: 277895

Peak memory: 27.99 MB

Total allocated: 113.99 MB

Total freed: 107.66 MB

Unique threads: 1

Duration: 1.93 seconds

Loading stats log from: peak_stats-p496734.csv

Stats graphs saved to: peak_stats-p496734_graphs.png

Library usage report saved to: peak_stats-p496734_library_report.txt

=== Stats Log Summary ===

Total functions profiled: 10

Total profiled time: 0.03 seconds

Total exclusive time: 0.03 seconds

Total overhead: 0.0132 seconds

Total calls: 9500

=== Top 10 Functions by Total Time ===

function	total_s	exclusive_s	count
zhpev_	0.010592	0.010592	26
zgemm_	0.009319	0.009319	2410
zcopy_	0.002355	0.002355	1634
dznrm2_	0.001615	0.001615	1414
dgetrf_	0.001491	0.001491	10

ddot_	0.001044	0.001044	2574
zdotc_	0.000873	0.000873	596
dscal_	0.000856	0.000856	804
zhemm_	0.000854	0.000854	22
dgetri_	0.000548	0.000548	10

=== Library Usage Summary ===

Library	Calls	% Calls	Time (s)
BLAS	9454	99.89	0.0169
LAPACK	10	0.11	0.0015

=== BLAS Breakdown ===

L1-Vector	7022	74.28%
L3-Matrix-Matrix	2432	25.72%

ABINIT's 0th test seems to very heavily on BLAS for matrix and vector operations, specifically ddot for vector dot products and zgemm for matrix multiplication

Methodology

1. Build the program
2. Wrote a script to run the program
3. Ran under idev
4. After successful test run, I ran under PEAK with memory, BLAS, LAPACK, and FFTW profiling enabled
5. Did some basic analysis and visualization of the outputs using a python script

Created a python script to do basic analysis profiling outputs

I created a python script to visualize the outputs of PEAK. It graphs both stat and mem csv's as well as data about target group libraries.

peak_analysis_v2.py

Sorted the top 100 list for actionable and unique items

Removed generic names, interpreters, and consolidated names. Also had AI write up a blurb for each item so I have a better idea of the programs use and operations.

Filtered top 100 list Filtering Report

Put together a general research data gathering plan / guideline / methodology for myself

1. Build the program

2. Use the ABINIT script as a base for a run script
3. Test run under idev
4. Enable PEAK profiling for memory, BLAS, LAPACK, and FFTW
5. Use another script to generate and submit different SLURM jobs in order to test affects of Node and MPI scaling. Multiple test inputs will also be run.
6. Use the python script to organize and visualize the data, grouped by the different scalings

Developed a workflow for myself

- I'm familiar with git so I'll be using a git repo on both the remote and local to keep my data, notes, and scripts organized and available to myself and the group.

What I learned this week

- building on HPC systems
- how to run PEAK on HPC systems and profile for BLAS, LAPACK, and FFTW
- I'm a bit more familiar with the top 100 items now

Questions / Issues

- Does my proposed data gathering plan sound good?
- Does the data gathered from ABINIT seem correct?

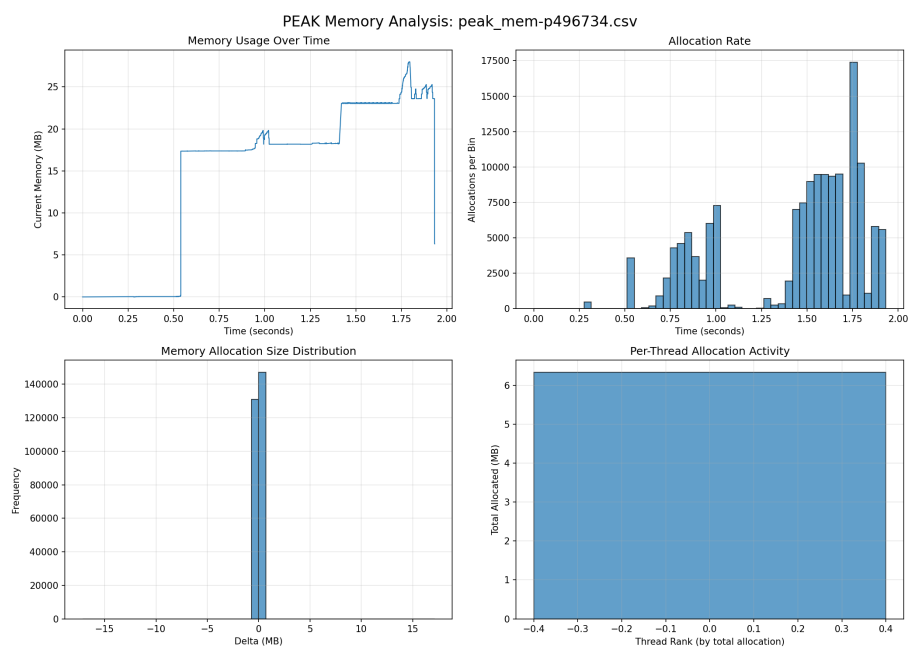


Figure 1: Memory Graph

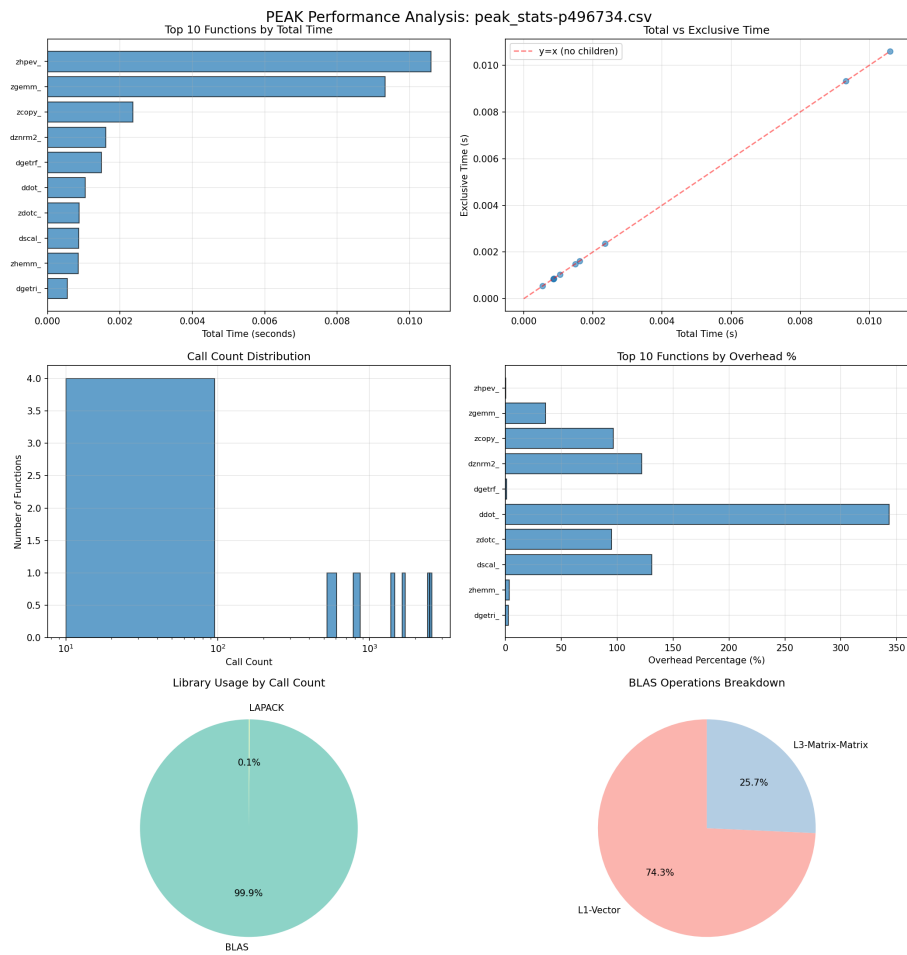


Figure 2: Stat Graph